



Executive Summary: Student Exam Scores Analysis

Overview

This analysis investigates how students' **weekly study hours** impact their performance across three key subjects — **Maths**, **Physics**, and **Chemistry**. The study uses exploratory data analysis techniques including **group-wise aggregations**, **visualizations**, and **correlation analysis** to derive actionable insights.

Key Insights & Highlights

1. Study Hours and Academic Performance

Students were grouped based on the number of hours they studied weekly:

Study Hours Category	% of Students	Avg. Maths Score	Avg. Physics Score	Avg. Chemistry Score
0–5 hours	31.2%	~52	~54	~51
6–10 hours	43.7%	~65	~67	~62
11+ hours	25.1%	~79	~76	~75

- Nearly **69%** of students studied **more than 5 hours per week**, showing a significant commitment to learning.
- Students who studied **11+ hours** scored approximately:
 - **52% higher in Maths**
 - **41% higher in Physics**
 - **47% higher in Chemistry** compared to those studying **0–5 hours**.

Conclusion: More study time **strongly correlates** with better performance.

2. Subject-Wise Score Comparison

- **Average scores** across all students:
 - **Maths:** ~65 (highest)
 - **Physics:** ~64
 - **Chemistry:** ~62 (lowest)

While performance is relatively strong in all subjects, **Chemistry** lags slightly, suggesting it may be the most challenging or neglected.

3. Heatmap of Subject Correlation

- A strong **positive correlation** exists between scores in all three subjects.
 - Maths vs Physics: **0.91**
 - Maths vs Chemistry: **0.89**
 - Physics vs Chemistry: **0.88**
- This indicates that students performing well in one subject often perform well in others, suggesting consistent study habits or general aptitude.

 This opens opportunities for **integrated or cross-subject study techniques**.

Recommendations

1. **Encourage a Minimum of 6–10 Study Hours Weekly:**
 - Students in this group consistently outperform the lower group by **20–30%** in scores.
2. **Targeted Interventions in Chemistry:**
 - Despite high correlations, Chemistry scores are marginally lower across all study hour groups.
3. **Support Cross-Subject Skill Development:**
 - Leverage the inter-subject correlation to design interdisciplinary teaching aids.
4. **Monitor and Mentor Low-Study Groups:**

- The 0–5 hour group (~31.2%) shows **high failure risk**, needing attention through personalized mentoring.

Final Thoughts

Your analysis clearly demonstrates that:

- **Time invested in studying = tangible improvement** in performance.
- Visualization-driven storytelling is effective in uncovering and communicating trends.
- Encouraging a **balanced and consistent study routine** can yield up to **50% performance gains** across subjects.